

```
mysql> use Practice1;
Database changed
mysql> create table roles(
  -> role_id
  -> int primary key,
  -> role_name
  -> varchar(50)
  -> );
Query OK, 0 rows affected (0.10 sec)

mysql> create table users(
  -> user_id
  -> int primary key,
  -> name
  -> varchar(50)
  -> email
  -> varchar(50);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'e
mail
varchar(50)' at line 6
mysql> create table users(
  -> user_id
  -> int primary key,
  -> name
  -> varchar(50),
  -> email
  -> varchar(50),
  -> role_id
  -> int
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> create table departments(
  -> dept_id
  -> int primary key,
  -> dept_name
  -> varchar(50)
  -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create table departments(  
  -> dept_id  
  -> int primary key,  
  -> dept_name  
  -> varchar(50)  
  -> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> create table employees(  
  -> emp_id  
  -> int primary key,  
  -> emp_name  
  -> varchar(50),  
  -> dept_id  
  -> int,  
  -> manager_id  
  -> int,  
  -> salary  
  -> decimal(10,2)  
  -> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> create table customers(  
  -> customers_id  
  -> int primary key,  
  -> customers_name  
  -> varchar(50),  
  -> city  
  -> varchar(50)  
  -> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> create table orders(  
  -> order_id  
  -> int primary key,  
  -> customer_id  
  -> int,  
  -> order_date  
  -> data,  
  -> total_amount  
  -> decimal(10,2)  
  -> );
```

```
  -> decimal(10,2)  
  -> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> create table accounts(  
  -> account_no  
  -> int primary key,  
  -> holder_name  
  -> varchar(50),  
  -> balance decimal(10,2)  
  -> );
```

Query OK, 0 rows affected (0.06 sec)

Empty Set (0.00 sec)

mysql> SHOW TABLES;

Tables_in_practice1
accounts
customers
departments
employees
orders
roles
users

7 rows in set (0.05 sec)

mysql> INSERT INTO departments VALUES(1,'CSE'),(2,'ECE');

Query OK, 2 rows affected (0.13 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO employees VALUES(101,'Ramesh',1,NULL,60000),(102,'Suresh',1,101,55000),(103,'Mahesh',2,101,45000);

Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO roles VALUES(1,'Admin'),(2,'Student');

Query OK, 2 rows affected (0.23 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO users VALUES(1,'Anita','anita@mail.com',1),(2,'Rahul','rahul@mail.com',NULL);

Query OK, 2 rows affected (0.22 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO customers VALUES(1,'Kiran','Hyderabad'),(2,'Meena','Delhi');

Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name,email,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;

name	email	role_name
Anita	anita@mail.com	Admin
Rahul	rahul@mail.com	NULL
NULL	NULL	Student

3 rows in set (0.01 sec)

mysql> SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.emp_id;

Employee	Manager
Ramesh	NULL
Suresh	Ramesh
Mahesh	Ramesh

3 rows in set (0.00 sec)

mysql> SELECT o.order_id, c.customers_name, c.city, o.total_amount
-> FROM customers c
-> JOIN orders o ON c.customers_id = o.customer_id;

order_id	customers_name	city	total_amount
201	Kiran	Hyderabad	3000.00
202	Meena	Delhi	5000.00

2 rows in set (0.00 sec)

mysql> SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;

dept_name	COUNT(*)	AVG(salary)	MAX(salary)
CSE	2	57500.000000	60000.00
ECE	1	45000.000000	45000.00

2 rows in set (0.02 sec)

mysql> SELECT emp_name,salary,dept_name FROM employees e JOIN departments d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='CSE';

emp_name	salary	dept_name
Ramesh	60000.00	CSE
Suresh	55000.00	CSE

2 rows in set (0.01 sec)

```
mysql> SELECT customer_name,SUM(total_amount) TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;
ERROR 1054 (42S22): Unknown column 'customer_name' in 'field list'
mysql> SELECT c.customers_name, SUM(o.total_amount) AS TotalPurchase
-> FROM customers c JOIN orders o ON c.customers_id = o.customer_id GROUP BY c.customers_name ORDER BY TotalPurchase DESC;
+-----+-----+
| customers_name | TotalPurchase |
+-----+-----+
| Meena          | 5000.00      |
| Kiran          | 3000.00      |
+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
ERROR 1146 (42S02): Table 'practice1.students' doesn't exist
mysql> SELECT s.student_name, c.course_name, f.faculty_name FROM students s JOIN enrollments e ON s.student_id = e.student_id JOIN courses c ON e.course_id = c.course_id JOIN faculty f ON c.faculty_id = f.faculty_id;
ERROR 1146 (42S02): Table 'practice1.students' doesn't exist
mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
ERROR 1146 (42S02): Table 'practice1.students' doesn't exist
mysql> SELECT s.student_name, c.course_name, f.faculty_name
-> FROM students s
-> JOIN enrollments e ON s.student_id = e.student_id
-> JOIN courses c ON e.course_id = c.course_id
-> JOIN faculty f ON c.faculty_id = f.faculty_id;
ERROR 1146 (42S02): Table 'practice1.students' doesn't exist
mysql> CREATE TABLE students (
-> student_id INT PRIMARY KEY,
-> student_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> CREATE TABLE courses (
-> course_id INT PRIMARY KEY,
-> course_name VARCHAR(50),
-> faculty_id INT
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> SELECT s.student_name, c.course_name, f.faculty_name
-> FROM students s
-> JOIN enrollments e ON s.student_id = e.student_id
-> JOIN courses c ON e.course_id = c.course_id
-> JOIN faculty f ON c.faculty_id = f.faculty_id;
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Ravi         | DBMS        | Dr. Sharma   |
| Meena        | Networks    | Prof. Reddy  |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> |
```